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The Relapse Prediction

Division-Count Decay of Stem Cell Rescue by Young Cell Transplantation as a Falsifiable Test of the Centriolar Damage Accumulation Theory of Aging

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Abstract

The transplantation of young stem cells into aged or progeroid organisms reliably extends lifespan and improves healthspan, yet this rescue is invariably partial and eventually wanes. Prevailing theories of aging—telomeric attrition, reactive oxygen species (ROS) accumulation, and epigenetic drift—predict that such interventions, particularly in syngeneic, immune-compatible systems, should yield stable or permanent rejuvenation of the host tissue niche. The observed transient benefit presents a fundamental paradox. Here, we propose a critical and falsifiable prediction derived from the Centriolar Damage Accumulation Theory of Aging (CDATA). CDATA posits that aging is driven by the progressive, post-translational polyglutamylation of mother centrioles, impairing ciliary signaling and ultimately disrupting cell division. This theory uniquely predicts that rescue by young cells, mediated via paracrine supplementation of Hedgehog/Wnt ligands, must be temporary. Crucially, relapse kinetics should be proportional to the post-transplant division count of the aged host cells, not calendar time. We formalize this "Relapse Prediction" (P11), detail an inexpensive, *in vitro* co-culture experiment to test it using aged hematopoietic stem cells (LSK population), and compare its implications against competing theories. Confirmation would establish centriolar damage as a primary cell-division clock and explain the universal limitation of stem cell therapies. Falsification would necessitate significant revision of CDATA, underscoring the predictive value of the framework.

Keywords: aging, stem cells, centriole, polyglutamylation, relapse prediction, TTLL6, CASIN, CDATA, serial transplantation

1. Introduction

The capacity of young stem cells to rejuvenate aged organisms is a cornerstone of modern geroscience. Seminal studies demonstrate that transplantation of young hematopoietic stem cells (HSCs) into unirradiated, aged mice extends median lifespan by up to 34% (Kovina et al., 2013), while muscle-derived stem cells (MDSPCs) can triple lifespan in a progeroid mouse model (Lavasani et al., 2012). Pharmacologically inhibiting CDC42 activity (CASIN) to "rejuvenate" aged HSCs similarly improves healthspan and lifespan (Leins et al., 2022). In humans, allogeneic bone marrow transplantation (BMT) for hematologic disorders often confers a youthful immune phenotype, temporarily reducing immunosenescence and frailty metrics.

However, this rescue is neither complete nor permanent. The degree of lifespan extension correlates with, but is not exceeded by, the level of chimerism achieved (Kovina et al., 2019). Clinically, the benefits of BMT on non-hematopoietic aging phenotypes eventually decline. Most tellingly, even young HSCs have a finite replicative capacity, exhausted after 4–5 serial transplantations in mice (Harrison & Astle, 1982). This indicates an intrinsic limit to cellular rescue that is independent of immune rejection, as these serial transplants are conducted in syngeneic settings.

No dominant theory of aging adequately explains why syngeneic rescue must be partial and temporary. Telomere theory is confounded by active telomerase in most transplanted stem cells. The ROS theory suggests young cells would permanently ameliorate the oxidative microenvironment. Epigenetic clock theories imply that paracrine factors from young cells could stably reprogram the aged epigenome. The observed decay of benefit challenges these models.

The Centriolar Damage Accumulation Theory of Aging (CDATA) offers a mechanistic, quantitative explanation (Tkemaladze, 2023, 2026). CDATA proposes that the mother centriole, which templates the primary cilium, accumulates damage via the progressive enzymatic addition of polyglutamate chains (polyglutamylation) by enzymes of the TTLL family, particularly TTLL6 in hematopoietic tissue (Whitfield et al., 2016; Mahalingan et al., 2024). This polyglutamylation — an irreversible post-translational modification (PTM) — disrupts the docking of key signaling complexes (e.g., for Hedgehog and Wnt pathways), leading to ciliary dysfunction, defective cell polarity, and eventual mitotic failure. Critically, this PTM accumulates with each cell division, as the mother centriole is asymmetrically retained (Yamashita et al., 2007; Cheng & Yamashita, 2008). CDATA thus frames aging not as a time-dependent process, but as a **division-counted decay of centriolar fidelity**.

From this framework, we derive a novel and non-obvious prediction: the rejuvenating effect of young stem cells on aged neighbors will decay in proportion to the number of divisions undergone by the aged cells post-rescue. We term this the **Relapse Prediction (P11)**. This paper formalizes P11, designs a definitive experimental test, and explores its profound implications for the biology of aging and the design of rejuvenation therapies.

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2. The Relapse Prediction: Mechanism and Mathematical Formulation

2.1. Mechanism of Rescue and Relapse

According to CDATA, aged cells suffer from impaired ciliary signaling due to excessive polyglutamylation on the mother centriole, reducing the output of key morphogens like Sonic Hedgehog (Shh) and Wnt. Young cells, with low centriolar PTM burdens, secrete these ligands at normal levels. When co-cultured or co-transplanted, young cells provide paracrine ligand supplementation (Florian et al., 2020), temporarily restoring functional signaling levels in the aged niche. This explains the immediate recovery in proliferation ($Ki67^+$) and function observed post-transplant.

However, relapse is inevitable. The aged cell's centriolar damage is not repaired; the underlying PTM load remains. The enzyme TTLL6 continues its activity during each cell cycle. Recent structural work has demonstrated that TTLL6 exhibits a **positive feedback loop**: it is stimulated by glutamylation of the β-tail of the laterally adjacent tubulin dimer, generating localized microtubule glutamylation patterns (Mahalingan et al., 2024). With every subsequent division, the mother centriole in the aged cell accrues additional polyglutamate chains. After N divisions post-rescue, the cumulative PTM exceeds a critical threshold where ciliary docking and signaling fail completely, regardless of ambient ligand concentration. At this point, the paracrine rescue from young cells becomes ineffective, and the aged cell returns to a senescent or dysfunctional state. **The system relapses.**

2.2. Mathematical Formulation

The dynamics can be modeled with a simple ordinary differential equation. Let P(t) represent the polyglutamylation load on the mother centriole of an aged cell. Its rate of change is proportional to the division rate k_div and the activity of TTLL6 (α), but is opposed by the minimal activity of deglutamylases like CCP1/AGBL5 (δ), which declines with age:

$$dP/dt = \alpha \cdot k_div - \delta \cdot P$$

For aged cells, $\delta \approx 0$, simplifying to $dP/dt = \alpha \cdot k_div$. The polyglutamylation load increases linearly with number of divisions:

$$P(N) = P_0 + \alpha \cdot N$$

where P_0 is the initial load at the time of young cell introduction. Rescue is effective while $P(N) < P_crit$, where P_crit is the threshold for ciliary failure. The number of divisions to relapse, N_relapse, is therefore:

$$N_relapse = (P_crit - P_0) / \alpha$$

Core prediction: The duration of rescue is a linear function of the number of post-intervention divisions of the aged cell, not of chronological time. If aged cells are held in deep quiescence, the rescue should persist indefinitely. If division is pharmacologically accelerated, relapse should occur more rapidly.

2.3. Quantitative Prediction

Based on known HSC division rates (approximately 1 per 30–50 days in steady-state) and the observed timeframe of functional decline in serial transplantation experiments (Harrison & Astle, 1982), **Cell-DT simulations** (Cell-DT, 2024; DOI: 10.5281/zenodo.19174506) — which model the accumulation of centriolar polyglutamylation as a function of division count — predict a rescue "half-life" corresponding to approximately **40–60 divisions** of the aged HSC post-co-culture initiation. Full relapse, defined as return to baseline dysfunctional proliferation and cilia metrics, is predicted at **80–120 divisions**. This range is testable *in vitro* where division cycles can be tracked and controlled using fluorescent division tracking dyes such as CellTrace Violet.

3. Comparison with Competing Theories

The Relapse Prediction provides a sharp discriminant between CDATA and other major aging theories, as summarized in Table 1.

Table 1: Predicted Rescue Duration Under Competing Theories

Theory	Predicted Rescue Duration (Syngeneic)	Outcome if Division-Counted Relapse Observed
Telomeric Attrition	Permanent (telomerase compensates)	**Falsified**
ROS/Damage Accumulation	Permanent (environment normalized)	**Falsified**
Epigenetic Drift	Stable (epigenome reprogrammed)	**Falsified**
Proteostatic Collapse	Unclear/Stable (chaperone rescue)	**Challenged**
CDATA (P11)	**Temporary — proportional to host division count**	**Confirmed**

The key distinction is the *cause* of relapse. Other theories might accommodate *time*-dependent relapse due to gradual re-accumulation of damage in young cells themselves. Only CDATA predicts a strict dependence on the *division count* of the *rescued aged host cells*. This makes the experimental design definitive.

4. Experimental Design

4.1. Cell Culture and Co-Culture System

Cells: Lin$^-$ Sca-1$^+$ c-Kit$^+$ (LSK) HSCs will be FACS-sorted from young (2–3 month) and aged (20–24 month) C57BL/6 mice. Young LSKs will be obtained from Act-GFP transgenic mice (GFP$^+$) to enable differentiation from aged GFP$^-$ cells.

Conditions:

1. **Old Alone:** Aged LSKs (GFP$^-$) cultured in standard HSC medium
2. **Young Alone:** Young LSKs (GFP$^+$) cultured in standard HSC medium
3. **Co-culture:** Aged (GFP$^-$) + Young (GFP$^+$) at 1:1 ratio
4. **Conditioned Media (CM):** Aged LSKs in media conditioned by young LSKs (72h, changed every 48h) — controls for paracrine vs. contact-dependent effects

Division Counting: CellTrace Violet (CTV) labeling will be performed at culture initiation. Two proliferation arms will be established:

- *Slow-cycling arm:* 1% FBS (minimal stimulation)
- *Fast-cycling arm:* SCF + TPO + Flt3-L supplementation (maximal stimulation)

Positive Control for TTLL6 Involvement: To confirm that polyglutamylation drives the observed relapse, parallel co-cultures will be treated with:

- **siRNA against TTLL6** (10 nM, 48h pre-treatment; Biorbyt orb1851294) (Mahalingan et al., 2024)
- **LDC10** (10 μM), a selective TTLL inhibitor (Al Kabbani et al., 2024)

Successful TTLL6 inhibition will be confirmed by reduced GT335 immunofluorescence intensity.

4.2. Sampling and Endpoints

At T=0 (baseline), +20 divisions, +60 divisions, and +100 divisions (tracked by CTV dilution), GFP$^-$ (aged) cells will be re-sorted for analysis of the following endpoints:

- **Proliferation:** Percentage of Ki67$^+$ cells (flow cytometry)
- **Centriolar PTM:** GT335 antibody intensity (polyglutamylated tubulin; immunofluorescence + flow cytometry)
- **Ciliary Integrity:** Percentage of ARL13B$^+$ cells and cilium length (immunofluorescence)
- **Functional Output:** Colony-forming unit (CFU) assays in methylcellulose

TTLL6 Expression Measurement: To assess whether TTLL6 expression differs between young and aged HSCs, we will perform:

- **qPCR:** TaqMan assay Mm01259160_m1 for TTLL6 mRNA
- **Western blot:** Anti-TTLL6 antibody (Abcam, ab203089)

4.3. Predictions and Falsification Criteria

CDATA P11 Prediction:

- Ki67 and CFU recovery in GFP^- cells will peak at T=20 divisions and decay by T=100 divisions
- Decay will correlate with division count (CTV dilution), not with culture time (slow-cycling arm will maintain rescue longer in clock time but relapse at similar division count as fast-cycling arm)
- GT335 signal will increase linearly with division count
- TTLL6 knockdown or LDC10 treatment will delay relapse (increase N_relapse)

Falsification of CDATA: If aged cells maintain full Ki67 positivity and functional rescue at +100 divisions **AND** show no increase in GT335 signal → the core CDATA postulate that division-counted PTM accumulation drives functional decline is falsified.

4.4. Power, Cost, and Feasibility

- n = 6 technical replicates per group
- Detects 30% difference in Ki67 recovery ($\alpha = 0.05$, power = 0.85)
- Estimated cost: **approximately $8,000**
- All techniques are standard in HSC biology; mouse strains are readily available

5. Therapeutic Implications

5.1. Re-defining Stem Cell Therapy Schedules

If P11 is confirmed, repeated reinfusions of young HSCs would be required every 5–10 years to maintain rejuvenation benefit. Single-dose "youth banking" strategies would fail to confer permanent benefit to the aged host niche. This aligns with the empirical ceiling observed by Kovina et al. (2019), where lifespan extension was proportional to chimerism fraction.

5.2. The CCP1/AGBL5 Axis as a Combinatorial Target

CDATA predicts that pre-treating aged HSCs with the deglutamylase CCP1 (clearing polyglutamylation prior to transplantation) would produce a 2–3× longer rescue duration, as P_0 in N_relapse = (P_crit − P_0)/ α would be minimized. Neither CCP1 nor AGBL5 appear in any of the 12 Hallmarks of Aging (López-Otín et al., 2023). No existing anti-aging screen identifies them. This is a CDATA-unique therapeutic axis.

5.3. Synergy with Reprogramming

Partial epigenetic reprogramming (OSKM) addresses downstream transcriptional consequences of ciliary dysfunction. Combined with upstream CCP1-mediated PTM clearance, effects would be multiplicative — repairing the structural lesion and resetting its transcriptional fallout simultaneously.

6. Conclusion

The Relapse Prediction (P11) is the most direct, testable, and distinctive claim of CDATA. It emerges naturally from the core premise: that an irreversible, division-counted post-translational modification drives the loss of cellular function. By predicting inevitable, division-dependent decay of young stem cell rescue in syngeneic settings, P11 starkly contrasts with all other major theories of aging.

The proposed experiment is simple (approximately $8,000), uses standard techniques, and is definitively falsifiable with the inclusion of:

- TTLL6 knockdown (siRNA) and pharmacological inhibition (LDC10) as positive controls
- qPCR and western blot measurement of TTLL6 expression in young vs. aged HSCs
- CellTrace Violet-based division counting to separate division number from chronological time

Confirmation of P11 would provide strong evidence that centriolar polyglutamylation is a fundamental mitotic clock governing stem cell aging across tissues and organisms. Falsification would necessitate significant revision of CDATA — in itself a scientifically valuable outcome that would redirect the field toward alternative mechanisms of asymmetric stem cell aging.

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